Solution

MOTION-04

Class 09 - Science

1. **(a)** Both A and R are true and R is the correct explanation of A.
   Explanation: Since velocity is a vector quantity, hence as its direction changes keeping magnitude constant, velocity is said to be changed. But for constant speed in equal time interval distance travelled should be equal.
2. **(c)** A is true but R is false.
   Explanation: A body has a uniform acceleration if it travels in a straight line and its velocity increases by equal amounts in equal intervals of time.
3. **(b)** Both A and R are true but R is not the correct explanation of A.
   Explanation: Uniform velocity means that speed and direction remain unchanged.
4. **(b)** Both A and R are true but R is not the correct explanation of A.
   Explanation: Average velocity V_{av}
   $= \frac{\text{Total displacement}}{\text{Total elapsed time}} = \frac{Vt}{t}$ = V
   = Instantaneous velocity
   Hence, assertion is correct. If a particle is in a round trip on a straight line, then average velocity is zero but at the instant at which the particle reverses its direction of motion, velocity is zero. So, reason is correct. But reason is not the correct explanation of assertion.
5. **(a)** Both A and R are true and R is the correct explanation of A.
   Explanation: Satellites revolve around their planets in almost circular orbits with constant speed. Thus, during their motion, the speed remains constant, while the direction of motion changes continuously. As a result, there is a change in their velocity. Therefore, the motion of satellites around their planets is considered as accelerated motion.
6. **(a)** Both A and R are true and R is the correct explanation of A.
   Explanation: This is the equation of a parabola. A parabola results when one quantity is proportional to the square of the other. When an object is moving with a constant velocity (zero acceleration), the displacement versus time graph is a straight line and its slope is velocity.
   When an object is moving at a constant speed but its direction of motion changes, its velocity changes ant thus acceleration is produced. The motion of an object in a circular path is such an example. In a uniform circular motion, the direction of motion of the object changes continuously and hence the velocity changes continuously even though the speed is constant.
7. **(b)** Both A and R are true but R is not the correct explanation of A.
   Explanation: Both A and R are true but R is not the correct explanation of A.
8. **(b)** Both A and R are true but R is not the correct explanation of A.
   Explanation: Displacement may be positive, negative or zero. Displacement is a vector quantity.
9. **(d)** A is false but R is true.
   Explanation: A boy goes from A to B with a velocity of 20 m/min and comes back from B to A with a velocity of 30 m/min. The average velocity of the boy during the whole journey is 24m/min.
10. **(b)** Both A and R are true but R is not the correct explanation of A.
    Explanation: The speedometer of a car measures the instantaneous speed of the car.
11. **(d)** A is false but R is true.

Explanation: As velocity is a vector quantity, its value changes with change in direction. Therefore when a bus takes a turn from north to east its velocity will also change.

12. **(a)** Both A and R are true and R is the correct explanation of A.

    Explanation: The uniform motion only means that the object is moving at a constant speed but its direction of motion may be changing at in the case of uniform circular motion. Hence, acceleration is produced in uniform motion due to changes in velocity.

13.

    (d) A is false but R is true.

    Explanation: Speedometer measures instantaneous speed of automobile.

14.

    (d) A is false but R is true.

    Explanation: Speed can never be negative because it is a scalar quantity.

15.

    (c) A is true but R is false.

    Explanation: When a particle is in uniform motion, the magnitude of its velocity at each instant such as t = 0, t = 1s, t = 2s ... etc. is always constant. Hence the velocity versus time graph for a particle in uniform motion along a linear path is a straight line parallel to the time axis.

16.

    (c) A is true but R is false.

    Explanation: The displacement is the shortest distance between initial and final position. When final position of a body coincides with its initial position, displacement is zero, but the distance travelled is not zero.

17. **(a)** Both A and R are true and R is the correct explanation of A.

    Explanation: According to statement of reason, as the graph is a straight line, $P \propto Q$, or, $P = \text{constant} \times Q$

    $\frac{P}{Q}$ = constant

    Equation of a straight line is y = mx + c

18. **(a)** Both A and R are true and R is the correct explanation of A.

    Explanation: A body has a non-uniform acceleration if its velocity increases by unequal amounts in equal intervals of time.

19.

    (b) Both A and R are true but R is not the correct explanation of A.

    Explanation: In this case, the bus travels 250 km from Delhi to Jaipur towards the West and then comes back to starting point Delhi in the reverse direction. So, the total displacement.

20.

    (b) Both A and R are true but R is not the correct explanation of A.

    Explanation: A body has a uniform speed if it travels equal distances in equal intervals of time, no matter how small these time intervals may be. For example, a car is said to have a uniform speed of say, 60 km per hour, if it travels 30 km every half hour, 15 km every quarter of an hour, 1 km every minute, and 1/60 km every second

21.

    (b) Both A and R are true but R is not the correct explanation of A.

    Explanation: When a body does not cover equal distances in equal intervals of time, the velocity is said to be variable or non-uniform velocity. In this case, the speed of the body is not constant. Even if the speed of a body is constant but the direction is changing, the velocity will not be uniform.

22. **(a)** Both A and R are true and R is the correct explanation of A.

    Explanation: Initial velocity (u) = 0 , acceleration (a) = 4 m/s^2

    v = u + at

    $v = 0 + 4 \times 10$

    v = 40 m/s

23.

    (d) A is false but R is true.

    Explanation: If the position-time graph of a body moving uniformly in a straight line is parallel to the position axis, it means that the position of body is changing at constant time. The statement is abrupt and shows that the velocity of body is infinite.

24. State True or False:

(i) **(a)** True
Explanation: True

(ii) **(a)** True
Explanation: True

(iii) **(a)** True
Explanation: True

(iv) **(a)** True
Explanation: True

(v) **(a)** True
Explanation: True

(vi) **(a)** True
Explanation: True

(vii) **(a)** True
Explanation: True

(viii) **(a)** True
Explanation: True

(ix) **(a)** True
Explanation: True

(x) **(a)** True
Explanation: True

(xi) **(b)** False
Explanation: False. The speed of a body is a scalar quantity. It can be zero or positive but can never be negative.

(xii) **(a)** True
Explanation: True

(xiii) **(a)** True
Explanation: True

(xiv) **(b)** False
Explanation: False. Speed of a body is defined as the distance travelled by the body in unit time.

(xv) **(a)** True
Explanation: True

(xvi) **(a)** True
Explanation: True

(xvii) **(a)** True
Explanation: True

(xviii) **(b)** False
Explanation: False. Displacement is a vector quantity.

(xix) **(a)** True
Explanation: True

(xx) **(a)** True
Explanation: True

25. Fill in the blanks:

(i) 1. Circular

(ii) 1. Motion

(iii) 1. Vector

(iv) 1. Non-uniform

(v) 1. Retardation

(vi) 1. Scalar

(vii) 1. Positive

(viii) 1. Uniform acceleration

(ix) 1. Displacement

(x) 1. Circumference

(xi) 1. Acceleration

(xii) 1. Zero

(xiii) 1. Speed

(xiv) 1. Same

(xv) 1. Origin

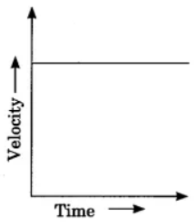
(xvi) 1. Velocity

(xvii) 1. Acceleration

26. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).
27. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i).
28. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i).
29. The correct order of match is given as (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii).
30. The correct order of match is given as (a) – (iii), (b) – (iii), (c) – (iv), (d) – (i).
31. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).
32. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i).
33. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).
34. The correct order of match is given as (a) – (ii), (b) – (iii), (c) - (iv), (d) – (i).
35. The correct order of match is given as (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii).

36. i. (c) the acceleration
 ii. (d) Velocity of a moving body is its speed in a given direction
 iii. (b) non-uniformly accelerated motion
 iv. (b) uniform acceleration
 v. (d) Displacement- Body in uniform motion

37. i. (a) 24 m/min
 ii. (a)



 iii. (c) equal to one
 iv. (a) (I) and (II)
 v. (c) has some initial velocity

38. i. (c) Motion of a racing car on a circular track
 ii. (d) Accelerated motion
 iii. (a) 5.5 m/s
 iv. (d) (II) and (IV)
 v. (c) Direction in which the sprinter is running

39. i. (a) Uniform speed
 ii. (d (III) and (IV)
 iii. (b) m^2
 iv. (d) 28m
 v. (c) The distance travelled by scooter is the minimum in this section

40. i. (a) 8m
 ii. (b) Car B is the slowest
 iii. (b) uniform acceleration
 iv. (a) (I), (II), and (III)

v. (a)